# PAPER_718: Red Dwarf Compression C: LENR, Higgs Field, Pi/Phi Series, NGC 346

**Class:** `UQFFKnowledgeBaseKB3` **CP4 Entry:** #302 **Keywords:** UQFF, LENR, Higgs, weak interaction, NGC 346, Pi series, Phi golden ratio, buoyancy, neutron production **Session:** 176 | **Version:** v5.33 **Source:** grok_share_ba508f76c8e.txt

## Abstract

Doc 43.c (Red Dwarf Compression_C) integrates electro-weak LENR theory with UQFF, includes the Higgs field mapping $m_H = 125$ GeV, derives Pi and phi series influences, evaluates NGC 346 gas nebula star formation ($T \approx 1.424 \times 10^6$ K), and connects the buoyancy framework (volume ratio $V_{little}/V_{big} = 1/33$) to UQFF vacuum densities.

## 1. LENR Electro-Weak Reaction

$$W + e^- + p \to n + \nu_e \qquad Q \approx 0.78\,\mathrm{MeV}$$

Neutron production rate $\eta$ in UQFF:

$$\eta = k_\eta \cdot e^{-[SSq]^{n_{26}} \cdot e^{-\pi - t}} \cdot \frac{U_m}{\rho_{UA}}$$

with $k_\eta = 10^{-113}$ (calibrated), $[SSq] = 0.57$, $n_{26} = 26$.

## 2. Higgs Field Integration

$$U_H = \lambda_H \cdot \rho_{UA}^{\prime:SCm} \cdot \omega_H \cdot e^{-[SSq]^{n_{26}} \cdot e^{-\pi - t}} \cdot (1 + f_{quasi})$$

Higgs mass calibration: $m_H = 125$ GeV $\implies k_{Higgs} = 1.79 \times 10^{18}$.

## 3. Pi and Phi Series Influences

$$\text{Pi influence} \propto U_m \cdot \pi \cdot \rho_{UA} \cdot \cos(\omega_c t)$$
$$\text{Phi influence} \propto U_m \cdot \phi \cdot \rho_{UA}$$
$$\text{FSC influence} = \alpha_{fine} \cdot U_m \quad \text{where} \quad \alpha_{fine} = \frac{1}{137}$$

with $\omega_c = 2\pi/(3.96 \times 10^8) \approx 1.585 \times 10^{-8}$ rad/s.

## 4. NGC 346 Star Formation

$$T_{starformation} \approx 1.424 \times 10^6\,\mathrm{K}$$

Blueshift velocity: $v_{blue} \approx -3.33 \times 10^{-5}\,c$

## 5. UQFF Buoyancy Equation

$$g_{buoy} = \frac{\rho_{UA}}{\rho_{SCm}} \cdot \frac{V_{little}}{V_{big}} = 10 \cdot \frac{1}{33} \approx 0.303$$

## References


- UQFF Framework Doc 43.c (Red_Dwarf_Compression_C)
- grok_share_ba508f76c8e.txt entry #67
- Session 176, v5.33


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